Secure UTM VM Setup

Complete Step-by-Step Guide for Apple M Series Mac's

100% Isolated • Agent-Safe • Clone-Protected

Security Overview

This guide configures UTM with six security layers that together fully isolate the VM from your Mac. An agent given full access to the VM cannot read, write, or delete anything on your Mac — not through the filesystem, not through the network.

Security Layer	Where Configured	What It Prevents
No shared folders	Disabled in UTM settings	Agent cannot read or delete any Mac files
Emulated VLAN network	Selected in UTM network settings	Agent cannot reach Mac or home LAN over network
Clone as backup	Right-click VM in UTM	Full VM backup — restore if anything goes wrong
UFW firewall	Enabled inside VM	Blocks all inbound connections to the VM
SSH server off	Disabled inside VM	No remote access pathway into the VM
Sudo password required	Configured inside VM	Agent cannot silently escalate privileges

ℹ Why UTM over Lima?

Lima's defaults favour sharing files with your Mac. UTM's defaults favour isolation. You are working with UTM's design rather than against it, which means fewer things to manually lock down and fewer ways to accidentally misconfigure.

Step 1 — Install UTM

Download UTM directly from the official website. It is free and open source.

Go to https://mac.getutm.app, download the latest release, and drag UTM into your Applications folder.

Step 2 — Download Ubuntu Server ARM64

You must use the Ubuntu Server ARM64 image. This is a lightweight, stable base — no bloated desktop packages — which is the correct foundation for a secure VM.

Download from: https://ubuntu.com/download/server

- Scroll to the section labelled 'Manual server installation'
- Download the ARM64 image — it will be named ubuntu-xx.xx-live-server-arm64.iso

> ⚠ **Always use the ARM64 image — never AMD64**
> The AMD64 image requires full CPU emulation on M series Macs and is significantly slower. The ARM64 image runs natively on M series Mac via Apple's Hypervisor Framework. If you accidentally download the AMD64 image, delete it and download again.

Step 3 — Create the VM in UTM

Open UTM and follow each screen exactly as described below. Every setting matters for isolation.

3a — Virtualization type

- Click "+" in UTM
- Select "Virtualize" — NOT "Emulate"

Virtualize uses Apple's native Hypervisor Framework for maximum performance. Emulate uses software emulation which is slow and unnecessary.

3b — Operating system

- Select Linux

3c — Boot ISO

- Click Browse and select the ubuntu server arm64 ISO you downloaded

```
Memory:     4096 MB  (use 8192 if your Mac has 16GB+ RAM)
CPU Cores:  4
Storage:    40 GB
```

3d — Shared Directory — leave completely empty

> ⚠ **Critical isolation step — do not skip**
> Leave the Shared Directory field completely blank. Click Continue without selecting any folder. This is the single most important setting in the entire guide. If you add a shared directory here, the agent inside the VM can read, write and delete your Mac files directly.

3e — Name the VM

```
Name: secure-ubuntu
```

- Click Save

Step 4 — Harden the Network

Before booting, verify the network mode. Click the VM in UTM, then click Edit (the pencil icon), then select Network.

```
Network Mode:  Emulated VLAN     ← select this
```

Emulated VLAN gives the VM internet access through NAT but prevents it from reaching your Mac or any device on your local network. Traffic goes out to the internet, but nothing can route back to the host Mac.

> ⚠ **Never use these network modes**
> Bridged (physical) — VM joins your real LAN and can reach your Mac. Host Only — VM can directly reach your Mac. Shared Network — behaves like Bridged in UTM. Only Emulated VLAN is safe for this setup.

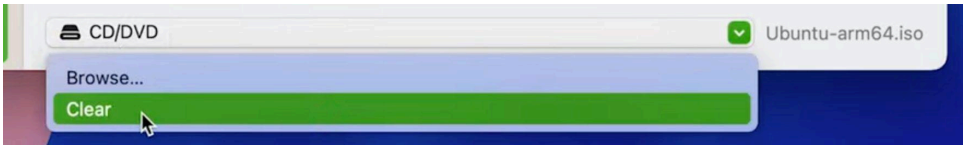
Step 5 — Boot and Install Ubuntu Server

1. Click the Play button in UTM to boot the VM
2. Ubuntu Server installer loads — wait for the menu to appear
3. Select "Try or Install Ubuntu Server" and press Enter
4. Follow the installer with these exact settings:

```
Language:           English
Keyboard:           Your layout
Installation type:  Ubuntu Server (not minimized)
Network:            Leave as default — DHCP will configure automatically
Storage:            Use entire disk (this is the virtual disk only, not your Mac)
Username:           vmuser
Server name:        secure-vm
Password:           Use a strong password, minimum 16 characters
```

5. Confirm the storage changes and let installation complete — approximately 10 minutes
6. When prompted, select Reboot Now

7. **IMPORTANT — Clear the ISO image in the UTM window immediately after reboot prompt.** Click on the VM in UTM, open the CD/DVD drive dropdown, and select **Clear**. If you do not do this, every time you start the VM it will boot back into the installer instead of your installed system.



> ⚠ **Do not skip the ISO clear step**
> This is the most commonly missed step when setting up UTM. If the ISO is not cleared, UTM will boot from the installer image every single time, and you will be stuck in a loop thinking the VM is broken.

Step 6 — Harden the VM

Log into the VM using the username and password you set during installation. Run each block below in order.

6a — Lock down sudo

```
echo 'Defaults timestamp_timeout=0' | sudo tee /etc/sudoers.d/timeout
```

```
sudo chmod 440 /etc/sudoers.d/timeout
```

This requires a password for every sudo command with no time-based caching. An agent cannot silently run privileged commands.

6b — Enable firewall

```
sudo ufw enable
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw status verbose
```

Expected output: `Status: active` with incoming DENIED

6c — Disable SSH server

```
sudo systemctl disable ssh
sudo systemctl stop ssh
sudo ufw deny 22
```

SSH is not needed since you interact with the VM directly through UTM. Disabling it removes any remote access pathway entirely.

6d — Verify no Mac filesystem is mounted

```
mount | grep -E 'virtiofs|9p|host|mac'
```

Expected output: nothing at all. Silence confirms the VM has no access to your Mac filesystem.

6e — Verify network isolation

```
# Should NOT be able to reach the UTM host gateway
ping -c 3 192.168.64.1
```

Expected: `Destination Host Unreachable` or `Request timeout`

```
# SHOULD be able to reach the internet via NAT
ping -c 3 8.8.8.8
```

Expected: replies received — internet works through NAT

Step 7 — Clone the VM as Your Backup

UTM does not have a built-in snapshot feature. Instead, use the Clone function to create a full copy of the VM at its clean, hardened state. This clone is your backup — if the agent ever damages the VM, you delete the damaged one and work from the clone.

How to clone

8. Stop the VM completely
9. Right-click the VM in the UTM sidebar
10. Select "Clone"
11. Name the clone: secure-ubuntu-BACKUP
12. UTM creates a full, independent copy of the entire VM

> ✓ **You are now fully set up**
> The VM is isolated, hardened, and backed up via clone. You can hand an agent full access to the original VM. Even if the agent deletes everything inside it, your Mac is completely untouched and you can restore from the clone instantly.

If the VM is ever damaged

13. Delete the damaged VM in UTM
14. Right-click the backup clone
15. Clone it again to create a fresh working VM
16. Rename the new clone back to secure-ubuntu
17. You are back to a clean state — your Mac was never at risk

> i **Keep the backup clone untouched**
> Never run the backup clone as your working VM. It exists solely as a restore source. Always work in the original VM and restore from the clone when needed.

Verification Checklist

Run these checks inside the VM at any time to confirm isolation is intact. Open a terminal and run each command. Every result must match the expected output — if any does not, refer to the relevant step above to fix it.

Check	Command (run inside VM)	Expected Result
Shared folders	`mount | grep -E 'virtiofs|9p|host'`	No output at all
SSH server	`systemctl is-active ssh`	inactive
Firewall	`sudo ufw status`	Status: active
Host reachable?	`ping -c 2 -W 1 192.168.64.1`	Request timeout or Unreachable
Internet via NAT?	`ping -c 2 8.8.8.8`	Replies received

Quick Reference

UTM actions — on your Mac

- Start VM: click Play in UTM
- Stop VM: click Stop in UTM or run sudo shutdown now inside the VM
- Clone VM: right-click VM → Clone (do this before any risky session)
- Delete VM: right-click VM → Delete (your Mac files are never affected)

Commands — inside the VM

```
# Check firewall status
sudo ufw status

# Check no shared mounts exist
mount | grep -E 'virtiofs|9p|host'

# Shut down cleanly
sudo shutdown now
```

If something goes badly wrong inside the VM

18. Stop and delete the damaged VM in UTM
19. Clone the backup (secure-ubuntu-BACKUP) to create a fresh copy
20. Rename the new clone to secure-ubuntu
21. Done — your Mac was never involved

> ⚠ **The one rule above all others**
> Never add a Shared Directory in UTM settings. That single setting is the only direct pathway between the VM and your Mac files. No shared directory means no pathway — the agent can do anything it wants inside the VM and your Mac remains completely untouched.